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User Acceptance of Agile Information Systems: A Model and Empirical Test

WEIYIN HONG, JAMES Y.L. THONG, LEWIS C. CHASALOW, AND GURPREET DHILLON

WEIYIN HONG is an associate professor in the Department of Management Information Systems, University of Nevada, Las Vegas. She received a Ph.D. in information systems from the Hong Kong University of Science and Technology and a B.Sc. from Fudan University, China. Her research interests include user acceptance of emerging technologies, human–computer interaction, and user privacy concern. Her work has appeared in *Information Systems Research*, *Journal of Management Information Systems*, *International Journal of Human–Computer Studies*, *Communications of the ACM*, *Information & Management*, *Journal of the American Society for Information Science and Technology*, and *Journal of Database Management*.

JAMES Y.L. THONG is a professor of information systems at the HKUST Business School, Hong Kong University of Science and Technology. He received a Ph.D. in information systems from the National University of Singapore. His research on technology adoption, human–computer interaction, computer ethics, and IT in small business has appeared or is forthcoming in *Information Systems Research*, *Journal of Management Information Systems*, *MIS Quarterly*, *Journal of the AIS*, *Information Systems Journal*, and *European Journal of Information Systems*, among others. He has served or is serving as an associate editor for *Information Systems Research* and *MIS Quarterly*.

LEWIS C. CHASALOW is an assistant professor of business at the University of Findlay, Ohio. He received a Ph.D. in MIS from Virginia Commonwealth University and an MBA, M.Sc., and B.Sc. in industrial engineering from Lehigh University. He has more than 30 years of experience as a senior IT executive at Capital One Financial, IBM, and AT&T. His research interests are in business intelligence, information systems strategy, and systems development processes.

GURPREET DHILLON is a professor of information systems in the School of Business, Virginia Commonwealth University, Richmond, Virginia. He holds a Ph.D. from the London School of Economics and Political Science. His research interests include management of information security and ethical and legal implications of IT. His research has been published in *Information Systems Research*, *Journal of Management Information Systems*, *Information & Management*, *Communications of the ACM*, *Computers & Security*, *European Journal of Information Systems*, *Information Systems Journal*, and *International Journal of Information Management*. He has authored seven books, including *Principles of Information Systems Security: Text and Cases*, and his comments have appeared in *Knowledge@Wharton*, *New York Times*, *USA Today*, and *BusinessWeek*, and on *NBC News*. He is editor-in-chief of the *Journal of Information System Security*.

ABSTRACT: In response to the rapid changes in users' requirements, a new generation of information systems (IS), namely, agile IS, has emerged. Agile IS, defined as information systems developed using agile methods, are characterized by frequent upgrades with a small number of new features released periodically. The existing research on agile IS has mainly focused on the developers' perspective with little research into end users' responses to these agile IS. Drawing upon the tripartite model of attitude, the status quo and the omission bias theories, and the availability heuristic, we propose a model that utilizes constructs from the unified theory of acceptance and use of technology, the IS continuance model, habit, and individual differences to examine the drivers of user acceptance of agile IS. Further, we investigate not only users' intentions to continue using the agile IS but also their intentions to use new features when they are released, which is a surrogate for the ultimate success of agile IS. Data from 477 users of an agile IS showed that users' level of comfort with constant changes, the facilitating conditions provided, and users' habit are predictors of both types of intentions, with users' level of comfort with constant changes being the strongest predictor. Users' intentions to continue using agile IS are also determined by users' satisfaction with and perceived usefulness of the past upgrades. Finally, users who are innovative are more likely to use future releases of new features. The present work fills a gap in the software engineering literature and contributes a technology acceptance model specific to agile IS, which are becoming a mainstay of companies' IT portfolio in a fast-changing business environment.

KEY WORDS AND PHRASES: agile methods, agile systems, availability heuristic, comfort with change, habit, information systems continuance, omission bias, personal innovativeness, status quo bias, unified theory of acceptance and use of technology (UTAUT).

IN TODAY'S COMPETITIVE BUSINESS ENVIRONMENT, the rapid changes in users' requirements are creating a demand for faster software development and upgrade speed [6]. With traditional development methods, such as the waterfall method, users are required to accurately describe their needs at the beginning of a project. Programmers will then follow rigid steps (e.g., analysis, design, implementation, and test) to deliver the requested features at the end of a relatively long development cycle. However, in a turbulent business environment, such projects will fail to provide the most needed software due to rapid changes in users' requirements and the high cost of modifying software [7, 43]. This situation has led to the creation of a new generation of software development methods, namely, agile methods, such as Extreme Programming or XP, SCRUM, and Crystal. Agile methods refer to a collection of principles and techniques that emphasize early and continuous delivery of valuable software with embracement of constant changes in users' requirements [19, 75]. Agile methods break the long development cycle into many smaller cycles, each containing the same development steps. But the goal is not to deliver a complete system with all requested features at the end of a development cycle. In fact, there is no intention to provide the full set of features when the system is delivered to users (e.g., the XP method calls for the development team to produce the first delivery within weeks). The system is constantly

Table 1. Characteristics of Agile IS Versus Nonagile IS

	Agile IS	Nonagile IS
Applicable context	More fluid user requirements	Relatively stable user requirements
Identification of user requirements	Users are constantly solicited for new requirements; emphasis on adaptivity to changing environments	User requirements are typically identified at the start of the development cycle, with emphasis on planning and predicting
Number of development cycles	Many short development cycles	One long development cycle
Development steps within each development cycle	Rigid steps	Rigid steps
Functions available when system is first released	System only provides a limited set of functions when first released	System is expected to deliver a full set of functions when first released
Goal in each development cycle	Each release has limited scope, i.e., each release delivers only a few valuable functions	A major release that comes with a complete set of functions
Typical release frequency	Frequent; typically every few weeks to every few months	Infrequent; typically after a few years
Example systems	iPhone apps, company intranets, Web-based systems, software as a service, etc.	Operational systems, enterprise resource planning, office automation systems, etc.

growing by implementing only the smallest set of the most valuable functions, based on users' input, in each round of release [7]. As a result, systems that are developed using agile methods provide only a limited set of functions when they are first introduced to users, but they will evolve periodically (e.g., monthly) based on a scheduled release cycle of features to address users' fluid requirements.¹ In this paper, we define agile information systems (IS) as the systems developed using agile methods. Table 1 summarizes the characteristics of agile IS as compared to nonagile IS.

The existing research on agile IS has mainly focused on the technical aspects of agile methods from the developers' perspective (e.g., [2, 15, 61]). In the studies that investigated the implementation experience of agile IS, the focus has been on the software development teams or project managers [31, 34, 44, 90] that advanced our understanding of the implementation of agile methods and the characteristics of agile IS. However, there is a lack of empirical research into the perceptions of agile IS from the users' perspective [17]. Some scholars have noted that agile IS are often conceived and perceived differently by their constructors versus their users [32]. The phenomenon is described as a "double dream," which is not necessarily the same for

all “dreamers.” While both users and developers have their own expectations of an agile IS, their collaborative efforts may result in an artifact that only partly embodies their respective expectations. For example, the developers’ dream is to provide a few most needed functions in each round of release. Users, however, may expect all functions to be provided in one release. In addition, different users may have different ideas about the most needed functions. Even for the same function, different users may hold different expectations of how and when they should be implemented. Hence, it is important to examine the agile IS phenomenon from the users’ perspective, which has not received adequate attention in the literature.

The lack of empirical research into the perceptions of agile IS from the users’ perspective may also reflect an underlying assumption that users will willingly embrace the changes because they are designed to address users’ needs. This assumption may not hold, as dealing with constantly evolving agile IS is not effortless for users. The psychology literature suggests that humans have a tendency to prefer options that cause no change to the state of the world (i.e., status quo bias [63]) or require no action on their part (i.e., omission bias [58]). The status quo and the omission bias theories describe people’s reactions to changes, which are a defining characteristic of agile IS. Some researchers suggest that humans resist change because it often involves more work in the short term [39], or simply because of the uncertainty associated with the change [63]. Applying this notion to the context of agile IS, users may find it difficult to deal with the frequent changes to the system interface and functions. As a result, the majority of the users, who are recipients and observers of agile IS,² have to weigh the potential benefits enabled by the constant upgrades against their tendency to resist changes to the system interface and functions. Given these considerations, users’ acceptance of agile IS cannot be taken for granted, and this issue warrants in-depth research.

Given the unique characteristics of agile IS and the limited understanding of users’ reactions toward agile IS, this paper aims to investigate the drivers of user acceptance of agile IS. We address the following research questions:

RQ1: What are the factors that facilitate users’ willingness to use agile IS?

According to the postadoption literature, users’ willingness to continue using an IS is mainly driven by a consideration of the benefits, that is, whether the system provides useful and satisfactory functions [10]. The cost to continue using an IS is considered minimal, as the learning effort has already been made during the initial adoption decision. In the case of agile IS, cost considerations are relevant throughout system usage due to the frequent changes to the system interface and features. The postadoption literature also predicts that if the current set of functions fails to satisfy users’ expectations, then users will discontinue use of the IS. However, for agile IS, users may decide to continue using the system even if the existing functions are disappointing, in anticipation that future upgrades will improve the system. Hence, there may be a different set of predictors for users’ continuance intention of agile IS.

RQ2: What are the factors that make users willing to try new features enabled by frequent upgrades to agile IS?

This issue is particularly relevant for agile IS because their success ultimately depends on the capability to continuously provide new features to address users' fluid requirements, and on the assumption that users will actually use those features as they become available. If users ignore the new features or do not feel compelled to learn how to use the new features, then the benefits of developing agile IS will be greatly compromised. Suffice to say that the answers to these research questions can provide important theoretical and practical implications about agile IS to both researchers and practitioners alike.

Theoretical Background

Tripartite Model of Attitude

WE PROPOSE THAT THE TRIPARTITE MODEL OF ATTITUDE PROVIDES a comprehensive framework for studying user acceptance of agile IS. *Attitude* is defined as a response to an antecedent stimulus or attitude object [13]. Rooted in the trichotomy of knowing, feeling, and acting as the three main facets of human experience [45], attitude has been portrayed as a tripartite model with three major dimensions, that is, *cognitive*, *affective* (or *emotional*), and *behavioral*, in the psychology literature [25]. The cognitive dimension refers to an individual's beliefs, thoughts, and perceptual responses about the attitude object; the affective dimension refers to an individual's feelings, emotional responses, or gut reactions engendered by an attitude object; and the behavioral dimension reflects an individual's evaluations of an attitude object based on past behaviors [13, 57]. The tripartite model of attitude has been empirically validated by psychologists in various social behavior contexts, including those involving change (e.g., [13, 81, 92]).

Cognitive Dimension

Early attitude research tends to focus on the cognitive dimension of attitude. According to the most popular conceptualization of attitude, that is, the expectancy-value model, individuals form beliefs about an object based on evaluations of its attributes [28, 29]. Each belief associates the object with a certain attribute, and an individual's overall attitude toward an object is determined by the subjective values of the object's attributes in combination with the strength of the associations. The inherent assumption of the expectancy-value model of attitude is that evaluative judgments are the result of cognitive processes [3].

Affective Dimension

A significant body of psychology literature suggests that human behavior can be better understood if affective processes are taken into consideration in addition to salient beliefs (e.g., [26, 47, 48]). In *changing* environments, affect may even exert a stronger impact on human behavior as compared to cognition (e.g., [14, 38, 80]).

for the following reasons. First, from an evolutionary perspective, humans' ability to adapt to changing environments is critical for their survival. Affect can aid in this adaptive process by providing meanings to stimuli [38]. Second, affect is usually more accessible than cognitions [89]. In our daily lives, one can usually tell how one feels about a person or an object, even if one cannot readily explain why. Finally, the dual-process theories propose that reliance on affect and emotion is a quicker, easier, and more efficient way to cope with complex and changing environments than analytical reasoning and beliefs [16, 70]. Hence, affect is a critical dimension of attitude, and especially in a changing environment.

Behavioral Dimension

Besides cognition and affect, individuals may evaluate an object based on past behavior. The self-perception theory argues that people often use their own behavior as a reference for their attitude toward an object or an action [9], that is, "doing" leads to "liking." As a result, repeated or habitual behaviors can inform people about their attitudes. For example, the fact that a person frequently eats Japanese food will inform him or her about his or her positive attitude toward this cuisine. This inference occurs as long as the behavior is seen as voluntary [11].

In the management literature, the tripartite model of attitude has been used to examine employees' attitude toward organizational changes. For example, it has been successfully applied to understand employees' attitudes toward organizational and technological changes [27]. Piderit [57] used the tripartite model of attitude to examine employees' ambivalence attitudes toward an organizational change. She argues that resistance to organizational changes can be understood from a multidimensional view of attitude. Conceptualizing each dimension as a separate continuum allows for the possibility of different reactions along the different dimensions. For example, an individual's cognitive response to a proposed change could be in conflict with his or her emotional response (e.g., an individual may see the potential long-term benefits of a proposed change but be apprehensive of the increased workload in the short term). In sum, the tripartite model of attitude allows researchers to have a comprehensive view of individuals' attitudes toward change.

We believe that the tripartite model of attitude provides a useful platform to examine user acceptance of agile IS. The majority of the IS adoption literature has emphasized the cognitive dimension, as users are typically viewed as rational individuals who make adoption decisions based on their evaluations of system characteristics and environmental characteristics (e.g., [23, 36, 76, 88]). The affective dimension of attitude has been examined in the context of hedonic IS (e.g., perceived enjoyment is a predictor of usage intention [84]), or as antecedent of the cognitive dimension (e.g., computer anxiety, an emotion variable, affects usage intention through perceived ease of use [85]). Because agile IS are defined by changes [20] and the significance of affect in a changing environment [14, 38], the affective dimension of attitude will play a more central role in user acceptance of agile IS. Finally, as agile IS promote early delivery of valuable functions, users can start using the IS at an early stage. Follow-

ing the self-perception theory, users' past usage behavior can inform them about their attitudes toward the IS. Hence, through early delivery of the agile IS, users are given the opportunity to develop habitual usage behavior, which increases the possibility of sustaining usage of the agile IS.

Status Quo Bias, Omission Bias, and Availability Heuristic

While the tripartite model of attitude can be used to explain general attitude formation, combining it with the status quo and the omission bias theories and the availability heuristic will better capture the changing nature of agile IS. The psychology literature shows that humans have a tendency to prefer options that cause no change to the state of the world (i.e., status quo bias [63]) or require no action on their part (i.e., omission bias [58]). There are two main reasons why people prefer no change and no action. First, transaction cost can make switching from the status quo or performing new actions costly. Transaction cost is triggered whenever the cost of switching or performing a new action exceeds the potential gain associated with the change. Applying this theory to user acceptance of agile IS, if individuals perceive the cost of learning a new feature is higher than the potential benefit of using that feature, they will be reluctant to try it. Second, even when there is no explicit cost associated with the change, uncertainty itself can lead people to status quo inertia. As agile IS are constantly changing, it is difficult for users to predict how useful the new features will be, or how they will affect the existing functions and user interface. The uncertainty associated with the changes can lead people to prefer no change and no action. Hence, the status quo and the omission bias theories emphasize the importance of examining users' reactions to frequent changes with agile IS.

In addition, predicting intentions to continue using an agile system and to explore new features involve some uncertainty, as new features are based on dynamic users' requirements and there is no predefined development plan of what the "final" system will look like. The psychology literature suggests that people rely on simple heuristics to make judgments under uncertainty [67, 83]. Heuristics are cognitive rules that people use to simplify difficult judgments [91, p. 279]. According to the availability heuristic, people will use a subset of information or knowledge that comes to mind most easily when making judgments involving uncertainty. In the context of user acceptance of agile IS, we argue that when forming intentions to continue using a changing system or to use unknown future upgrades (both of which involve uncertainty to users), the information that comes to mind most easily is the perceptions of the past upgrades. These past upgrades not only define agile IS but also indicate how the IS will grow in the future. For example, assume that an agile system was introduced six months ago and it had gone through four upgrades so far. The current system was still less than optimal with many requested functions missing. But the users noticed that some useful features were added in the last few upgrades, which gave them confidence that the system was growing in a positive direction, and thus they were more willing to continue using the system and to explore new features that became available. Combining the focus on past changes with the tripartite model of attitude, we will examine

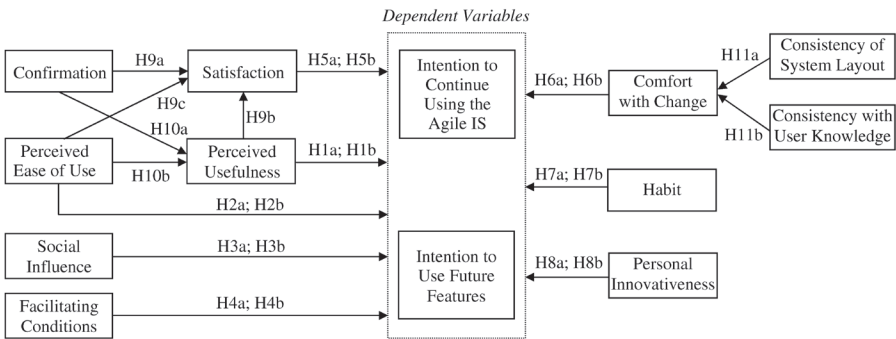


Figure 1. Research Model

how users’ cognitive beliefs, affective beliefs, and usage behavior of *past upgrades* (or changes) can affect their acceptance of agile IS.

Research Model and Hypotheses

WE DEVELOP OUR RESEARCH MODEL BY IDENTIFYING KEY CONSTRUCTS under the three dimensions of attitude (see Figure 1). There are two dependent variables of interest: intention to continue using the agile IS and intention to use future features of agile IS,³ that is, new features that are implemented in the system periodically. When identifying the antecedents, we focus on users’ beliefs, affects, and usage behavior of the *past upgrades*, and use the IS adoption literature to guide our effort, as building on the rich existing knowledge of IS adoption is important for knowledge accumulation. To complete our theoretical model, we incorporate an individual characteristic variable that is particularly relevant to individuals’ reactions to changes.

Cognitive Dimension

To identify the key cognitive variables, we reviewed the IS adoption and postadoption literature, which has mainly emphasized a cognitive evaluation of IS in determining usage intention. The IS adoption literature examines the situation where users with limited exposure to a system have to decide whether to adopt it [76, 86, 87, 88]. The postadoption literature focuses on the situation where users have already adopted the IS and have relatively rich experience with it. They are faced with the decision to continue or terminate the usage of the IS [10, 35, 77]. Both streams of research are relevant to the current context because agile IS, at any given time, have system features that fit the postadoption context (i.e., relatively “old” features that users have more experience with) and features that fit the IS adoption context (i.e., relatively “new” features that users have limited or no experience with). Hence, following the technology acceptance model (TAM) [22] and the unified theory of acceptance and use of technology (UTAUT) [88], which are two influential theories in the IS adop-

tion context, and the postadoption IS continuance model [10], which is an influential theory in the postadoption context, we identify perceived usefulness, perceived ease of use, social influence, and facilitating conditions as the key cognitive variables. We use these cognitive variables to predict not only the continuance intention but also intention to use future upgrades. The later dependent variable is unique to the agile IS adoption context and reflects the uncertainty involved in the cost and benefit evaluation of using new features (as the majority of potential users cannot be actively involved in the development of new features). In such an uncertain environment, people tend to rely on simple heuristics in making decisions [67, 83].

Perceived Usefulness

Here, perceived usefulness (PU) is defined as the extent to which users believe using the past upgrades of agile IS enhances their job performance (adapted from [88]). PU is generally accepted as an important predictor of behavioral intention in prior literature (e.g., [36, 78]), although in the IS adoption models, PU is generated from limited experience with the system, while in the postadoption IS continuance model [10], PU is based on extensive experience with the system. As an agile IS contains a mixture of “old” and “new” features at any given time, both models may apply, which suggests that PU will have a positive effect on intention to continue using the agile IS (H1a). Applying this notion to the understanding of intention to use future upgrades of agile IS, ideally, PU of future features will be a positive antecedent. However, as users do not have experience with using future features until they are implemented, it is not possible to evaluate their perceptions beforehand. Under such uncertainty, we argue that users will follow the availability heuristic, that is, users who perceive the past upgrades to be useful are more likely to hold similar beliefs about the usefulness of future releases of new features.⁴ Hence, we predict that PU of past upgrades has a positive effect on intention to use future features (H1b).

Hypothesis 1a: PU has a positive effect on intention to continue using the agile IS.

Hypothesis 1b: PU has a positive effect on intention to use future features.

Perceived Ease of Use

Here, perceived ease of use (PEOU) is defined as the extent to which users believe using the past upgrades of agile IS is free of effort (adapted from [88]). The IS adoption literature suggests that PEOU will have a direct effect on behavioral intention (e.g., [35, 77]). The argument is that when users have limited experience with the IS, ease of use of the system will translate to greater adoption intention as learning to use the system will require less effort. In the postadoption IS continuance context [10], however, PEOU is typically not included as an antecedent of behavioral intention. The assumption is that for IS that are already in use, the difficulty experienced during the initial learning stage will gradually disappear as users gain more experience

with the IS. In the agile IS context, as there are always new features introduced into the system, learning becomes a constant requirement. Moreover, the addition of new features may result in changes to the existing user interface, which requires additional learning effort. Thus, we predict that PEOU will have a positive effect on intention to continue using the agile IS (H2a). Following the same rationale, PEOU of future upgrades will have a positive effect on intention to use future upgrades. But, as such perceptions will be difficult to obtain before the future features are implemented, we argue that users will follow the availability heuristic, that is, users who find the past upgrades easy to use are more likely to assume that future releases of new features will similarly be easy to use. Thus, we predict that PEOU of past upgrades will have a positive effect on intention to use future features (H2b).

Hypothesis 2a: PEOU has a positive effect on intention to continue using the agile IS.

Hypothesis 2b: PEOU has a positive effect on intention to use future features.

Social Influence

Here, social influence (SI), an important social norm belief, is defined as the degree to which individuals perceive that important others believe they should use the past upgrades of agile IS (adapted from [88]). According to the IS adoption models, SI can influence behavioral intention. For behaviors conducted in a social context, other people's opinion can play a role in affecting one's intention to carry out the behavior. While SI is explicitly included in IS adoption models, it is typically not included in postadoption IS continuance models (e.g., [10]) where the assumption is when users first decide whether to adopt the IS, their decisions have already taken other people's opinions into consideration. Instead, subsequent decisions in the postadoption context will be mostly performance based. In the context of agile IS adoption, users repeatedly face evaluation-and-adoption decision cycles such that important others' opinions can have ongoing influence on users' behavioral intention to use the IS. Hence, we predict that SI will have a positive effect on intention to continue using the agile IS (H3a). Applying this notion to the understanding of intention to use future upgrades, as new features are gradually added to the agile IS and become part of the system, any effect of SI on promoting the use of the system will naturally include the use of future features. In addition, the availability heuristic suggests that if users perceive that important others encourage them to use prior upgrades, then they are likely to assume that important others will continue encouraging them to use future upgrades. Hence, we predict that SI will also have a positive effect on intention to use future features (H3b).

Hypothesis 3a: SI has a positive effect on intention to continue using the agile IS.

Hypothesis 3b: SI has a positive effect on intention to use future features.

Facilitating Conditions

Here, facilitating conditions (FC), an important behavioral control belief, is defined as the degree to which individuals believe that an organizational and technical infrastructure exists to support use of the past upgrades of agile IS (adapted from [88]). It is generally believed that when the organizational and technical infrastructure facilitates usage of the IS, people are more willing to use them. FC is not included in the IS postadoption models (e.g., [88]), as it is assumed that the support enabled by FC is taken into account in the initial adoption decision. In the agile IS context, as users need to constantly learn how to use new features, a more supportive organizational and technical infrastructure will provide a more sustainable environment for the continued usage of the IS. Thus, we expect that FC has a positive effect on intention to continue using the agile IS (H4a). In addition, following the availability heuristic, users who perceive that there are enough resources to support the use of prior upgrades are more likely to assume that resources will be available for the use of future features. With continuous organizational and technical support, users may expect fewer obstacles to using new features, and thus be more willing to use these features (H4b).

Hypothesis 4a: FC has a positive effect on intention to continue using the agile IS.

Hypothesis 4b: FC has a positive effect on intention to use future features.

Affective Dimension

Following the postadoption model, we identify satisfaction as a key affective variable in the study of user acceptance of agile IS. In the marketing literature, satisfaction is typically conceptualized as an affective state or overall emotional reaction to a service experience [52, 71]. In the postadoption literature, satisfaction is defined as users' overall experience of the IS usage [10, 18]. In this study, we are interested in studying satisfaction with the *past upgrades* of agile IS; hence, we define satisfaction as users' overall experience with the past upgrades of agile IS.

With affective considerations gaining more attention from researchers in recent years, there are a number of affective variables that have been examined in prior adoption research, such as perceived enjoyment [77, 82] and pleasure and arousal [41]. While many of these variables focus on the fun part of using IS, they are not designed to capture users' reactions to the changing nature of agile IS. Thus, in this study, we examine a new affective variable, that is, users' level of comfort with constant changes that describes users' emotional reaction to changes.

Satisfaction

The IS continuance model predicts that satisfaction is a key determinant of the intention to continue using IS [10, 77]. In the case of agile IS, the system is introduced to users before all the requested features are available, thus allowing users to start using

the system earlier. If users perceive the past upgrades as satisfactory, they are more likely to continue using the system. Hence, we predict that satisfaction has a positive effect on intention to continue using agile IS (H5a). In addition, as users do not have direct experience with the future release of the system, they will apply the availability heuristic to make inferences of the new release based on the available information that they have on past upgrades. As a result, users may use their level of satisfaction with past upgrades as a reference point for their satisfaction levels with future releases of new features. Hence, we predict that users who are satisfied with the past upgrades provided by the agile IS will be more willing to adopt future features (H5b).

Hypothesis 5a: Satisfaction has a positive effect on intention to continue using the agile IS.

Hypothesis 5b: Satisfaction has a positive effect on intention to use future features.

Comfort with Change

In the psychology literature, comfort is defined as feeling “at ease,” which is considered to be a positive affect [21, 68]. Comfort is an important affective variable in a dynamic adoption context. First, comfort has been explored in attachment theory [12], which proposes that comfort is associated with proximity to familiar persons or objects [24]. As agile IS involve frequent changes and introduction of new features, they can overload users’ ability to keep up with the IS, which in turn affect the comfort level that users experience when interacting with the IS. Second, comfort is a relatively enduring feeling that is generated after multiple interactions with a person or an object [62]. This enduring characteristic is particularly relevant to the current study as users build a sense of comfort or discomfort through multiple interactions with the agile IS.

In studies of organizational changes, comfort with change has been identified as an important factor distinguishing employees who are more favorable to organizational changes from those who are not [51]. In another study on Internet users, it was found that among elderly people, those who are more comfortable with change are more likely to adopt Internet technology [82]. IS researchers have paid limited attention to this construct in the IS adoption context, probably because the existing research has mainly focused on IS where constant change is not a prominent issue.

In the agile IS context, comfort can conceivably play an important role in determining user acceptance of such systems. A major characteristic of agile IS is the periodic changes to system features, and correspondingly, changes to the system interface. Users who are comfortable with change will take a more relaxed approach in encountering new features and interface changes [66]. As a result, they are more likely to continue using the agile IS (H6a). Furthermore, users who are comfortable with change are likely to be more receptive toward an evolving system. Prior literature shows that employees who are comfortable with a new technology will have a greater sense of efficacy and stronger beliefs in their ability to master the technology [5]. These users will be more

willing to use the new features offered by agile IS. The availability heuristic also suggests that users who are comfortable with prior upgrades are more likely to believe that they will continue to be comfortable with future upgrades. Thus, we predict that comfort has a positive effect on intention to use future features (H6b).

Hypothesis 6a: Comfort with change has a positive effect on intention to continue using the agile IS.

Hypothesis 6b: Comfort with change has a positive effect on intention to use future features.

Behavioral Dimension

The psychology research into the tripartite model of attitude encourages using verbal statements about past behavior to capture the behavioral dimension of attitude [13, 60], because it is an individual's self-perception of his or her past behavior that informs the individual about his or her attitude toward an object or an action. Habitual usage is such a self-reported behavioral variable that has gained attention from technology adoption researchers [42, 54].

Habit

In the IS literature, habit is defined as “the extent to which using a particular IS has become automatic in response to certain situations” [42, p. 711]. The psychology literature argues that a well-practiced behavior will repeat because the initiation and control processes have become automatic [55]. When using the agile IS has become a habit, continued usage can be expected as a matter of automated action. Hence, we predict that habit has a positive effect on intention to continue using the agile IS (H7a). In addition, as agile IS are defined by constant changes, users who have developed a habit of using the systems are more likely to find the incremental changes a natural attribute of agile IS. In such cases, habit can also be understood as the extent to which users' reactions to the incremental changes have become automatic. Hence, we predict that habit has a positive effect on intention to use future features (H7b).

Hypothesis 7a: Habit has a positive effect on intention to continue using the agile IS.

Hypothesis 7b: Habit has a positive effect on intention to use future features.

Individual Differences

Besides the cognitive, affective, and behavioral dimensions of attitude, individual differences can also exert influence on adoption behavior [1, 36, 76, 78]. In the IS adoption literature, individual differences refer to user factors that include traits, such as personality and demographic variables, as well as situational variables that

account for differences attributable to circumstances, such as experience and training [1]. Individual differences may be used most generally to be suggestive of any dissimilarity across individuals. For example, in any given population, some individuals are more receptive to change or more willing to try new things than others [46]. Such a characteristic is captured by personal innovativeness, which refers to an individual's willingness to change [37]. As personal innovativeness is highly relevant to the study of user response to changes in general, and to changes in the agile IS context in particular, we incorporate it into our research model.

Personal Innovativeness

In the IS literature, personal innovativeness is defined as the willingness of an individual to try new IS [1]. "Innovative" people are characterized as those who are early adopters of an innovation [59]. If we consider agile IS as composed of many small innovations, then an innovative person is more likely to welcome these innovations [77]. They will be more accommodating of the periodic changes of agile IS and thus more willing to continue using the IS. Hence, we predict that personal innovativeness has a positive effect on intention to continue using the agile IS (H8a). Moreover, users with higher levels of personal innovativeness will also be early adopters of the small innovations in each round of release. They are more likely to explore and use the new features of agile IS, as innovative people have higher levels of risk tolerance [1]. Exploring the new features of agile IS involves risks, such as the risk of spending excessive time and effort on learning, the risk of not deriving expected benefits from the new features, and the risk of being disappointed. Users with high personal innovativeness are more willing to take risks, and thus more willing to use the new features. Hence, we predict that personal innovativeness has a positive effect on intention to use future features of the agile IS (H8b):

Hypothesis 8a: Personal innovativeness has a positive effect on intention to continue using the agile IS.

Hypothesis 8b: Personal innovativeness has a positive effect on intention to use future features.

Determinants of Intermediate Variables

In addition to hypotheses related to the two main dependent variables, we propose additional hypotheses related to some of the antecedents of the two intention variables to gain a richer understanding of user acceptance of agile IS. We examine the determinants of three intermediate variables, that is, satisfaction, PU, and comfort with change.

Determinants of Satisfaction

According to the postadoption model, confirmation and PU are the two main predictors of satisfaction with IS. Confirmation is defined as the extent to which users' expectation

is confirmed [10]. In the case of agile IS, users' satisfaction is built on the experienced usefulness of past upgrades, as well as whether the performance of these upgrades meets their prior expectations. Thus, we predict that both confirmation and PU have positive effects on satisfaction (H9a and H9b). PEOU is not explicitly included in the postadoption model because of the assumption that the learning process is completed after formal adoption of the IS. However, in the case of agile IS, the learning process is ongoing due to the periodic introduction of new features. As learning is an important aspect of the ease of use [22], we believe that PEOU will play a more enduring role in determining users' satisfaction. Thus, we predict that PEOU has a positive effect on satisfaction (H9c).

Hypothesis 9a: Confirmation has a positive effect on satisfaction.

Hypothesis 9b: PU has a positive effect on satisfaction.

Hypothesis 9c: PEOU has a positive effect on satisfaction.

Determinants of Perceived Usefulness

According to the postadoption model, confirmation also has an indirect effect on satisfaction through PU [10]. After adopting a system, users want to avoid cognitive dissonance by matching their prior perceptions of the system with their observed perceptions. Confirmation of prior expectations will improve users' perceptions of the system, while disconfirmation will reduce such perceptions. This continues to be true in the agile IS context, where many rounds of such confirmation processes may take place with periodical upgrades. Thus, we predict that confirmation has a positive effect on PU of the agile IS (H10a). Meanwhile, according to TAM, PEOU has a direct effect on PU. The rationale is that if a system is easy to use, it helps to reveal the usefulness of the system to users. This rationale is particularly true of agile IS where new features need to be learned and evaluated for their usefulness. Hence, we predict that PEOU has a positive effect on PU (H10b).

Hypothesis 10a: Confirmation has a positive effect on PU.

Hypothesis 10b: PEOU has a positive effect on PU.

Determinants of Comfort with Change

We propose that interface consistency contributes to users' sense of comfort in coping with the constant changes of agile IS. Interface consistency has received extensive attention among researchers over the years (e.g., [64, 65]). Although its definition varies, it is generally agreed among researchers that consistency helps to reduce errors and improve performance [50]. There are two types of interface consistency [74]: one is display consistency, which focuses on the consistency of system layout; the other is cognitive consistency, which focuses on the consistency of the interface with what the user knows. Following this distinction, we propose that consistency of system

layout and consistency with user knowledge are key determinants of comfort. Here, consistency of system layout is defined as the consistency of the textual and graphical appearance or the visual characteristics of the system layout; and consistency with user knowledge is defined as the consistency of texts and images with users' knowledge and usage conventions.

A major reason why users feel uncomfortable with agile IS is the periodic changes to the interface. As new features are introduced, developers will add new buttons, menu commands, or hyperlinks (for Web systems), which change the system interface. Releases of new features may also involve rearranging the placement of existing features or modifying existing color scheme. These interface changes may cause inconsistency in both system layout and user knowledge. These two types of inconsistency are related but independent of each other. For example, adding a new portlet (i.e., hyperlinks grouped together under a heading) on a Web site will affect the consistency of system layout, but as long as the portlet is designed in the same style as previous portlets, it will not affect consistency with user knowledge. But if a different search mechanism is implemented (e.g., from exact search to fuzzy search with wildcard characters), it will affect consistency with user knowledge, but will not change the existing interface. We predict that both types of consistency will have positive effects on comfort (H11a, H11b), as they minimize the difficulties users need to overcome when changes are introduced. With a consistent layout, users can easily ignore the new features if they choose to, and comfortably continue with their existing practices. Also, when new features conform to users' existing knowledge, they can be easily understood and thereby cause minimal discomfort.

Hypothesis 11a: Consistency of system layout has a positive effect on comfort.

Hypothesis 11b: Consistency with user knowledge has a positive effect on comfort.

Methodology

Company X's Agile IS

DATA WERE COLLECTED FROM COMPANY X, an alias for a Fortune 500 company in the service industry. Its annual revenue is over \$10 billion, and it has about 20,000 employees globally. Company X had started a project to replace its internal Web portal with a new one that was developed using an agile method. The previous portal had a static interface, and mainly provided company news, hyperlinks to other internal Web sites, and limited internal search functions. The new portal provided a single login to access a wide variety of corporate systems, customized news contents, customized tools, improved search functions, and so on. This agile IS won the "10 Best Intranets" award from the Nielsen Norman Group.

When the new Web portal was launched, only some basic features were available. An initial survey conducted by the company showed a high level of dissatisfaction with the new portal, which led to the management team's decision to keep the old portal running in parallel for six months. System usage was voluntary, as users were

not required to use the new portal to do their jobs. For instance, they could access the various corporate systems directly without going through the new portal. In fact, the survey found that some users actually bookmarked the URLs of individual corporate systems, thus negating the need for the new portal.

After the initial launch, new features were added periodically based on users' prioritization of the most requested features. Due to the size of the company, it was not viable to collocate all potential users with the development team. Instead, management decided to solicit feedback on the important features from all potential users through periodic surveys, with two user representatives directly involved in the agile development process and collocating with the development team. A standard development cycle was six weeks, and each release contained anywhere from two to three relatively significant enhancements to as many as six minor changes. Updates on the forthcoming new features were publicized to employees through e-mail newsletters and news postings on the portal before the actual releases.

Data Collection Procedure and the Sample

We conducted our survey approximately a year after the new Web portal was launched. There were eight upgrades during this period. The timing for the survey was opportune for two reasons. First, the users had time to get over the initial "growing pains" of adjusting to a completely new portal, such as unfamiliarity with the new portal's interface design and getting used to logging in before using the portal. By allowing an adjustment period, users' responses would be more stabilized and more reliable. Second, the portal was still undergoing development and expansion, with many releases planned after the survey. Hence, the new portal was still an active agile system.

The survey data were collected approximately four weeks after a release. Flyers about the online survey were distributed to employees attending an event at the U.S. headquarters of Company X. Respondents were given a chance to win gift prizes in return for their participation. The online survey was conducted over 5 days and received 507 responses. After removing responses with incomplete data, we had 494 usable responses. As the company had strict privacy regulations, we could not collect demographic data about the respondents, except for their job titles. According to the information technology (IT) manager in charge of the company's data warehouse, our distribution of job titles was representative of the employees who had access to the Web portal. About 2 percent of the respondents were in the executive board of the company (i.e., executive directors and senior directors), 37.7 percent in middle management (i.e., directors, senior managers, and managers), and 60.3 percent were operational personnel (i.e., other job titles).

To test for nonresponse bias, we compared the means of all variables in the research model between early and late respondents. The overall multivariate analysis of variance (MANOVA) test was not significant ($p = 0.93$), as were all the individual tests on each variable. Hence, nonresponse bias did not appear to be a major concern in our study. We then examined the possibility that our survey respondents tend to be heavy users of the agile system. We compared usage duration and usage frequency between early and late respondents. The overall MANOVA test was not significant ($p = 0.186$), as

were the individual tests on each variable. Hence, our sample was not biased toward heavy users of the agile system. Finally, we examined whether our sample was biased toward highly motivated users (i.e., users with high scores for the intention constructs). We used both intention measures as dependent variables and the date of participation in the survey as the independent variable. If highly motivated users were more likely to participate in the survey, then we should find “high” intention users participating earlier. The overall MANOVA test was insignificant ($p = 0.80$), as were the individual tests on each of the intention constructs, indicating that our sample was not biased toward users with “high” intentions.

Measurement of Constructs

Wherever possible, existing scales with multiple items were used to measure the constructs (see Appendix Table A1). All the constructs were measured using seven-point Likert scales, with the exception of satisfaction, which was measured using seven-point semantic scales. When measuring the independent variables, we focused on users’ responses to changes, that is, the past upgrades, instead of the whole system, whenever applicable. We provided a list of the past upgrades to the users so that their responses would be sensitized to the past upgrades as a frame of reference in answering the questions. Satisfaction and confirmation were measured by following the work of Bhattacharjee [10] and adapted to the agile IS context. Items measuring PU, PEOU, SI, and FC were adapted from Venkatesh et al. [88] to suit the current context. We developed the measure of comfort by modifying Thurstone and Chave’s [79] affect measure with a focus on the feeling of comfort. The two interface consistency measures, consistency of system layout and consistency with user knowledge, were measured by items adopted from the Interface Consistency Testing Questionnaire [56]. Habit was measured by two items developed by Limayem et al. [42], and personal innovativeness was measured by three items developed by Agarwal and Prasad [1]. The first dependent variable, intention to continue using the agile IS, was measured by two items following Bhattacharjee [10]. The second dependent variable, intention to use future features, was measured by items developed by following Ajzen and Fishbein’s [4] suggestions and by modifying existing intention scales in IS adoption studies [88]. We presented the questionnaire to the manager of the Web portal development team and several developers to verify that our description of the features was accurate and the questions were relevant and understandable to users of the Web portal. Next, we conducted a pilot study of the questionnaire on 30 users. Based on their feedback, we made some minor rephrasing to some of the instructions.

Data Analysis

IN ORDER TO ENSURE THAT DATA USED FOR ANALYSIS were provided by users who had gone through at least one upgrade (so that the independent variables, which focus on perceptions of past upgrades, were meaningful to them), we removed responses from

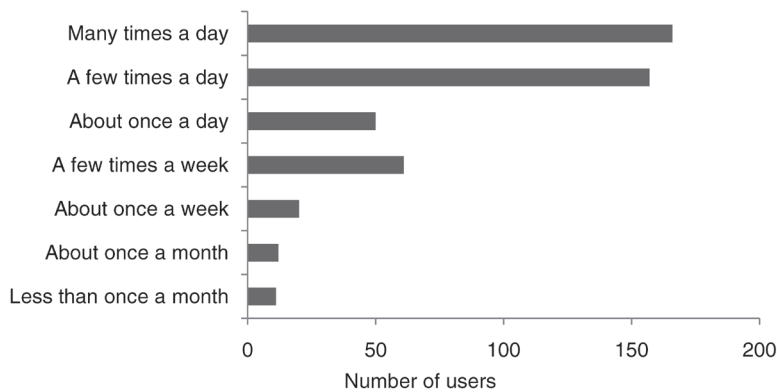


Figure 2. System Usage Frequency (Breadth of Usage)

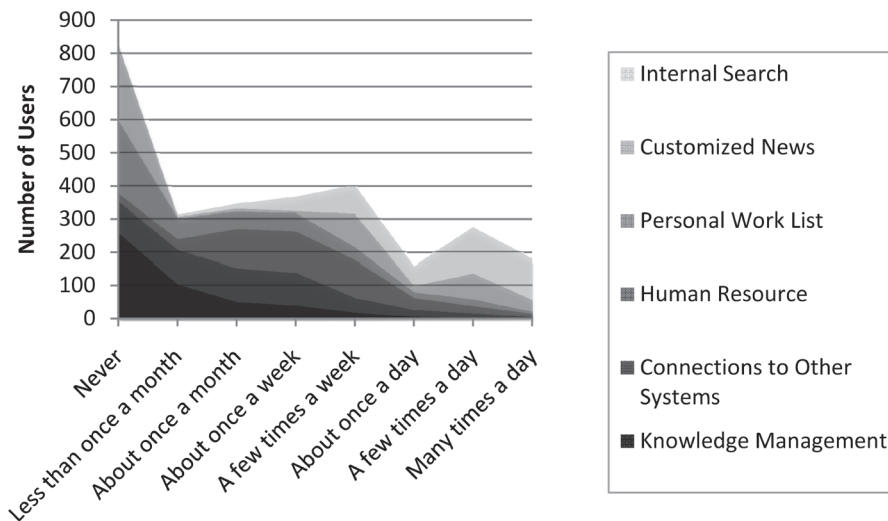


Figure 3. Functions Usage Frequency (Depth of Usage)

users who had less than four weeks of experience using the system (as the most recent upgrade was introduced four weeks prior to this survey). This left us with data from 477 respondents for further analysis. Figure 2 presents the usage frequency of the system (i.e., the breadth of usage) and Figure 3 shows the usage frequency of the six major functions of the system (i.e., the depth of usage) for these users. The six major functions of the system were internal search, customized news, personal work list, human resources, connections to other systems in the organization, and knowledge management. The system was heavily used by our respondents, with the majority of them using the system at least a few times a day. Not all of the functions were equally used; for example, the internal search function was used frequently, whereas the knowledge management function was used less frequently.

Instrument Validation

We used LISREL 8.7 to test the measurement model, which included all 13 constructs. The fit indices of the measurement model were all within the suggested ranges, indicating good model fit. The ratio of chi-square to degrees of freedom was 1.84 and within the suggested value of 3 [33]. The goodness-of-fit index (GFI) and adjusted goodness-of-fit index (AGFI) were 0.92 and 0.89, respectively. The normalized fit index (NFI), nonnormalized fit index (NNFI), and comparative fit index (CFI) were 0.98, 0.99, and 0.99, respectively, indicating good model fit. Finally, the root mean square residual (RMSR) provided an indication of the proportion of the variance not explained by the model, with an upper threshold value of 0.10. A similar index of root mean square error of approximation (RMSEA) described the discrepancy between the proposed model and the population covariance matrix, with an upper threshold value of 0.08. The RMSR and RMSEA for the measurement model were 0.03 and 0.04, respectively, well within the threshold values. Thus, we could proceed to evaluate the psychometric properties of the instrument.

We used Cronbach's alpha, composite reliability, and average variance extracted (AVE) to determine reliability and convergent validity of the constructs (see Tables 2 and 3). All the Cronbach's alphas and composite reliabilities were above the 0.70 threshold [33]. All the AVEs were at least 0.60, which was higher than the recommended value of 0.50 [33], meaning more than half of the variances observed in the items were accounted for by their hypothesized constructs. Convergent validity was assessed by the factor loadings and squared multiple correlations from the confirmatory factor analysis. All the factor loadings were greater than 0.70 [33], while squared multiple correlations between the individual items and their intended constructs were above 0.50 in all cases. Next, we examined discriminant validity by comparing the shared variances between constructs with the AVE of the individual constructs [30]. The shared variances between constructs were all lower than the AVE of the individual constructs, thus confirming discriminant validity. In sum, the measurement model showed adequate reliability, convergent validity, and discriminant validity. To test for potential common method bias, we performed a chi-square difference test between the one-factor model and our measurement model. The test showed a significant chi-square difference of 5,155.93 ($p < 0.01$). Thus, common method bias did not appear to be an inhibiting factor in this study.

Structural Model

All the fit indices of the structural model showed good fit for the research model.⁵ The ratio of chi-square to degrees of freedom was 2.16, within the recommended value of 3 [33]. Comparison of other fit indices with their corresponding recommended values also provided evidence of a good model fit (GFI = 0.90, AGFI = 0.87, NFI = 0.98, NNFI = 0.99, CFI = 0.99, RMSR = 0.06, RMSEA = 0.05). Hence, we could proceed to examine the path coefficients of the structural model (see Table 4). While PU had a significant positive effect on intention to continue using the agile IS ($\beta = 0.16$), it

Table 2. Descriptive Statistics and Convergent Validity

	Cronbach's alpha	Composite reliability	Mean	Standard deviation	Factor loadings	Squared multiple correlations
Intention to continue using the agile IS						
INTC1	0.73	0.75	6.12	0.84	0.82	0.68
INTC2			5.71	1.12	0.73	0.53
Intention to use future features						
INTF1	0.89	0.89	5.93	0.91	0.92	0.85
INTF2			5.83	0.93	0.87	0.76
Satisfaction						
SAT1	0.92	0.93	5.29	1.06	0.93	0.86
SAT2			5.28	1.10	0.92	0.85
SAT3			5.16	1.22	0.84	0.71
Confirmation						
CONF1	0.90	0.90	4.88	1.08	0.91	0.82
CONF2			4.82	1.09	0.89	0.80
Perceived usefulness (PU)						
PU1	0.94	0.94	5.11	1.18	0.92	0.84
PU2			5.03	1.17	0.94	0.88
PU3			5.25	1.12	0.89	0.79
Perceived ease of use (PEOU)						
PEOU1	0.92	0.92	5.42	1.11	0.91	0.83
PEOU2			5.26	1.13	0.86	0.74
PEOU3			5.44	1.06	0.91	0.83

(continues)

Table 2. Continued

	Cronbach's alpha	Composite reliability	Mean	Standard deviation	Factor loadings	Squared multiple correlations
Social influence (SI)	0.89	0.90				
SI1			4.26	1.47	0.89	0.79
SI2			4.34	1.33	0.92	0.84
Facilitating conditions (FC)	0.85	0.86				
FC1			5.37	1.17	0.78	0.61
FC2			5.39	1.20	0.95	0.90
Comfort with change	0.79	0.80				
COMF1			5.44	1.09	0.86	0.73
COMF2			5.75	1.03	0.77	0.59
Consistency of system layout	0.92	0.93				
CLAY1			5.60	0.93	0.90	0.81
CLAY2			5.67	0.85	0.90	0.81
CLAY3			5.64	0.93	0.89	0.79
Consistency with user knowledge	0.86	0.87				
CKW1			5.48	0.98	0.86	0.74
CKW2			5.48	0.94	0.90	0.82
Habit	0.92	0.92				
HAB1			5.75	1.24	0.90	0.82
HAB2			5.78	1.23	0.94	0.89
Personal innovativeness	0.91	0.91				
PI1			5.55	1.07	0.85	0.73
PI2			5.07	1.35	0.84	0.71
PI3			5.42	1.19	0.95	0.90

Note: Factor loadings are from confirmatory factor analysis.

Table 3. Discriminant Validity

Construct	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Intention to continue using the agile IS	5.91	0.89	0.60												
2. Intention to use future features	5.88	0.87	0.49	0.80											
3. Satisfaction	5.24	1.05	0.32	0.21	0.81										
4. Confirmation	4.85	1.03	0.22	0.15	0.42	0.81									
5. Perceived usefulness (PU)	5.13	1.09	0.32	0.21	0.53	0.58	0.87								
6. Perceived ease of use (PEOU)	5.37	1.02	0.31	0.27	0.55	0.45	0.56	0.80							
7. Social influence (SI)	4.30	1.33	0.12	0.12	0.26	0.35	0.35	0.29	0.82						
8. Facilitating conditions (FC)	5.38	1.10	0.24	0.21	0.21	0.18	0.22	0.45	0.21	0.76					
9. Comfort with change	5.59	0.96	0.42	0.36	0.49	0.35	0.56	0.64	0.19	0.24	0.67				
10. Consistency of system layout	5.63	0.84	0.25	0.28	0.31	0.28	0.30	0.42	0.13	0.20	0.52	0.80			
11. Consistency with user knowledge	5.48	0.91	0.28	0.26	0.34	0.30	0.40	0.50	0.15	0.26	0.53	0.59	0.77		
12. Habit	5.76	1.19	0.28	0.17	0.19	0.14	0.22	0.18	0.11	0.08	0.26	0.19	0.16	0.85	
13. Personal innovativeness	5.35	1.11	0.05	0.10	0.08	0.05	0.07	0.09	0.07	0.16	0.05	0.10	0.06	0.03	0.78

Notes: SD = standard deviation. Diagonals represent the average variance extracted. Other entries represent the shared variance.

had no effect on intention to use future features; H1a was supported while H1b was rejected. Both PEOU and SI had no significant effects on either intention to continue using the agile IS or intention to use future features; H2a, H2b, H3a, and H3b were not supported. FC had significant positive effects on both intention to continue using the agile IS ($\beta = 0.26$) and intention to use future features ($\beta = 0.16$), supporting H4a and H4b. Satisfaction had significant positive effects on intention to continue using the agile IS ($\beta = 0.19$), but no effect on intention to use future features; H5a was supported while H5b was rejected. Comfort was the strongest predictor of both intention to continue using the agile IS ($\beta = 0.30$) and intention to use future features ($\beta = 0.42$), supporting H6a and H6b. Habit had significant positive effects on both intention to continue using the agile IS ($\beta = 0.27$) and intention to use future features ($\beta = 0.15$), supporting H7a and H7b. Finally, personal innovativeness had a significant positive effect on intention to use future features ($\beta = 0.13$), but no effect on intention to continue using the agile IS; H8a was rejected while H8b was supported. Our model explained 51 percent and 43 percent of the variance in intention to continue using the agile IS and intention to use future features, respectively.

All other hypotheses with satisfaction, PU, and comfort as dependent variables were supported. Confirmation ($\beta = 0.15$), PU ($\beta = 0.29$), and PEOU ($\beta = 0.42$) had significant positive effects on satisfaction, confirming H9a, H9b, and H9c. The three variables explained 62 percent of the variance in satisfaction. Confirmation ($\beta = 0.47$) and PEOU ($\beta = 0.44$) had significant positive effects on PU and explained 70 percent of its variance, supporting H10a and H10b. Consistency of system layout ($\beta = 0.33$) and consistency with user knowledge ($\beta = 0.53$) were significant predictors of comfort, supporting H11a and H11b, and explained 66 percent of its variance.

Discussion

THERE ARE THREE MAIN FINDINGS FROM THIS STUDY. First, the tripartite model of attitude is supported by our data with the affective variable of comfort with change, the cognitive variable of FC, and the behavioral variable of habit as the three strongest predictors of both intentions. Second, there are clearly different drivers of intention to continue using agile IS versus intention to use new features, highlighting the significance of studying this new dependent variable. Third, the relatively high *R*-squares that we were able to achieve for both dependent variables (comparable to the results of prior IS adoption research) confirm the appropriateness of defining the independent variables as users' perceptions of past upgrades, instead of the whole system, in the context of constantly changing systems, such as agile IS.

More specifically, comfort with change, an affective component of attitude, was found to be the strongest predictor for intention to continue using the agile IS and also for intention to use future features. This result is consistent with the psychology literature, which suggests that affect often exerts dominating effects on human decision makers in turbulent environments. Users will use their feelings of how comfortable they are with system changes as the main heuristic to guide their decision making. For those who find the constant changes annoying or difficult to keep up with, they

will simply ignore the new features and stick to the existing functions that they are familiar with. In extreme cases, users may resent and even reject using the new interface if they are not comfortable with the changes. For example, Microsoft was forced to roll back a change to Windows Live Hotmail after users complained about its new interface release [40].

In addition, consistency of system layout and consistency with user knowledge are two important determinants of comfort. Our study suggests that if new releases bring significant changes to the layout of the existing system, such as moving existing hyperlinks or buttons to another location on the screen, then users will find it difficult to locate the existing features at their familiar positions, thus causing discomfort to the users. Similarly, if the new features are inconsistent with user knowledge or usage convention of the existing system, then users need to modify their established knowledge of the system to keep up with the changes or adjust the way in which they have been interacting with the system. For example, if a pen icon has been used to indicate customizable contents, then changing it to a hand icon will cause discomfort to the users, who now have to adjust their user knowledge.

FC is the strongest cognitive component of attitude that has significant effects on both intentions, suggesting that users of agile IS are sensitive to the “cost” of the adoption behavior. As long as the users perceive that adequate organizational and technical support is in place, which reduces the demand on effort by the users, they will be more willing to continue using the agile system as well as more willing to try new features.

Our study also shows a direct effect for habit, a behavioral component of attitude, on both intention variables. One benefit of agile IS is that they can “lure” people into using the systems before they become big and complex. Users can be overwhelmed by complex systems with a large number of features, which may intimidate them from using the systems. Agile IS, by contrast, can appear to be small and nimble at the beginning. Once users get used to the systems, they are less likely to stop using them even with periodic changes. Users who have developed a habit of using an agile system are also more likely to consider the constant changes as a natural part of the system, and thus more willing to try new features when released.

In addition to the above three common variables, we also found different drivers of intention to continue using agile IS versus intention to use new features. While PU and satisfaction are found to have positive effects on continuance intention, they have no effect on intention to use new features. The significant results are consistent with the postadoption literature, except that in the literature, only users who are satisfied with *the whole system* are likely to continue using the system, whereas in our study, finding *the past upgrades* useful and satisfactory provide significant motivation. These results confirm the utility of the status quo and the omission bias theories in studying constantly changing systems, as people’s attitudes are heavily influenced by their perceptions of change.

At the same time, there is support for the availability heuristic in predicting intention to use future features for some of the variables, such as comfort with change and

FC, but not others, such as satisfaction and PU. It appears that users place greater importance on using heuristics about their comfort level with the past upgrades and FC to guide their willingness to use future upgrades. The reason could be because it requires less effort to use these variables as heuristics, as compared to the other variables. While it does not require much effort in evaluating one's affective response to comfort, and FC provided by companies tends to be stable, users need to exert more effort to evaluate the ease of use and usefulness of prior upgrades.

Both PEOU and SI do not have significant direct effects on either of the two intentions variables. However, PEOU has significant effects on satisfaction and PU. In the postadoption literature, PEOU is not considered to be a significant predictor of satisfaction because it is typically assumed that users already know how to use the various features through their earlier interactions with the system. Similarly, in the IS adoption context, the effect of PEOU on PU is expected to decrease as users gain more experience with the system. However, in the agile IS context, learning is constantly needed due to the periodic release of new features. Hence, the ease with using system features continues to be a relevant factor in the adoption of agile IS. In the case of SI, psychologists (e.g., [81]) have argued that social norm is more important when the behavior is new, and becomes less critical when the behavior is routinized by the individual. As users become more exposed to an agile system, the effects of SI may diminish.

Finally, individual characteristics play a significant role in determining whether users intend to use future features. In particular, users with high levels of personal innovativeness are more likely to explore new features as they are more curious and more receptive of anything new. This finding suggests that some users are more willing to use new features than others, not necessarily because they think the future features are going to be beneficial to their work, but just because they like to try new things.

Implications and Limitations

Theoretical Implications

THIS STUDY MAKES FOUR IMPORTANT THEORETICAL CONTRIBUTIONS. First, to the best of our knowledge, this is one of the first attempts to empirically examine agile IS from the users' perspective and delineate the applicability of existing IS adoption and postadoption theories in the new context. Prior research on agile IS has focused on the technical aspects of agile methods from the developers' perspective. Our study fills the gap with an empirical investigation of agile IS from the users' perspective. As agile IS possess characteristics that can fit both the context of IS adoption (e.g., TAM and UTAUT) and postadoption (e.g., IS continuance model) theories simultaneously, it will be useful to investigate whether the existing adoption literature/models can be applied to understand acceptance of dynamic systems. By building on the rich existing knowledge of IS adoption, this study sets a baseline on how the existing models apply to the new context. Our results show that both types of theories are applicable

in understanding user acceptance of agile IS to some degree; however, neither of them can adequately explain users' reactions to agile IS individually. The current study has incorporated salient variables from both the IS adoption and postadoption literature, as well as identified new variables (e.g., comfort with change) to develop a more complete picture of user acceptance of agile IS.

Second, we have employed the tripartite model of attitude and demonstrated the importance of its cognitive, affective, and behavioral dimensions in understanding user acceptance of agile IS. The affective variable of comfort with change, the cognitive variable of FC, and the behavioral variable of habit are found to be the three most significant predictors for both intention to continue using agile IS and intention to use future features. Our research points to the need to evaluate all three dimensions of attitude in future research into agile IS acceptance. By incorporating the tripartite model of attitude, researchers will be able to gain a more complete understanding of the research phenomena.

Third, based on the status quo and the omission bias theories and the availability heuristic, we focused on studying users' reactions to changes (or past upgrades) instead of the whole system. In prior IS adoption research, the entire system was typically the target of study. But for agile IS, the system itself is dynamic and may mean different things to different users. For example, one user may consider the initial launch of the system disappointing but think highly of the subsequent upgrades while another user may like the initial system but consider the subsequent upgrades dissatisfying. If we had measured their perceptions of the whole system, it would be difficult to determine which part of the system they were referring to when giving the assessments. We argue that the periodic changes affect users' adoption intentions, following the status quo and the omission bias theories. These theories highlight the importance of users' perceptions of changes in affecting their attitudes and behavior. Moreover, the availability heuristic suggests that people will use a subset of information or knowledge that comes to mind most easily when making judgments involving uncertainty. As changes are ongoing for agile IS, the past upgrades will be prominent in users' minds when asked whether they would be interested in using future upgrades. In short, focusing on users' perceptions of changes opens up a new way to study dynamic systems, such as agile IS.

Finally, taking into account the IT artifact in the context of study [53], we identified a new dependent variable, *intention to use future features*, which is unique to the context of agile IS acceptance. This variable is a surrogate for the ultimate success of agile IS, as periodic releases of new features are ongoing and a defining characteristic of agile IS. Prior research has shown that there is a difference between initial use and intention to continue such use in the future [10]. We take it one step further by differentiating intention to continue using the existing system from intention to use new features. Our results show that there are different sets of drivers for the continuance intention variable and the new intention variable, suggesting that the factors that push users to continue using agile IS may not be adequate to facilitate utilization of the new features. In short, there is a need to delineate the theoretical distinction between continuance intention and intention to use future features when studying dynamic systems.

Practical Implications

This study provides important guidelines to companies and software developers who are developing agile IS, especially medium- to large-scale agile IS. Software developers should not assume that users will embrace the upgrades in each round of release. In other words, users will not necessarily use the new features just because they are using an agile system. Hence, software developers need to address the dual goal of not only retaining users but also encouraging users to explore the new features of the agile IS. We found that software developers need to pay special attention to the following issues when designing and developing upgrades so that users will be more likely to use them.

It is important to maintain the comfort levels of users when designing new upgrades. We found that comfort with change has a much stronger effect on users' intention to continue using the agile IS and their intention to use new features than the traditional enablers, such as PU or satisfaction, indicating that users put greater emphasis on the costs of adoption, in contrast to software developers, who focus on the benefits of the system. One strategy to attain a high comfort level is to maintain consistency in system layout so that the introduction of new features does not cause significant changes to the interface. For example, on a Web portal where all the features are accessible through hyperlinks, avoid moving the hyperlinks of existing features when adding new hyperlinks. Software developers should also ensure that the same color and font sizes are used for new hyperlinks and tabs so that they blend seamlessly into the existing interface. Another strategy is to ensure that the design of new system features is consistent with users' knowledge so that it takes minimal effort for users to learn how to use these new features. For example, the new Web portal in Company X has a number of portlets in the middle of the screen. The portlets that are customizable have a little button at the upper right corner, with a small pen icon on the face of the button, while the portlets that are not customizable do not have this icon. When developing new portlets, software developers should use the same icon for customizable portlets, as this knowledge has been learned by users through their past interactions with the system. These guidelines will ensure that users are comfortable with the periodic changes and will continue to use the system as well as explore future features.

The finding about comfort also has implications for nonagile IS when new versions are introduced as replacements for older versions of the software. For example, Microsoft Office 2007 is very different from Microsoft Office 2003, including a whole different look and feel, removal of all menus, changes to familiar toolbars, and different categorization and location of the buttons. As a result, many users have been hesitant to switch to the new version [69], even though the new design may be an improvement over the old design. The inconsistency in the system interface and inconsistency with users' existing knowledge post a steep learning curve even to seasoned users of the Office tools [49]. One can only imagine what would happen if such dramatic changes happened every few months or every few weeks. In short, ensuring users' comfort is imperative during software changes, and it is more critical for agile IS, where changes occur more frequently. With constant changes becoming increasingly

frequent in an era of smaller applications (e.g., iPhone and iPad apps), software as a service (SaaS), and Web-based systems, our findings have practical implications for these systems as well.

Besides ensuring the comfort level of users throughout the periodic releases of new features, there is another simple strategy that a company can adopt to facilitate the continued usage of agile IS, which is to promote the usage of the system at an early stage. Our results showed that once users have developed a habit of using an agile system, they will not only continue to use it but they will also be more willing to try new features. This is probably because once users have adapted to the constant changes, they view the changes as a natural part of the system. Hence, it is critical that companies promote the use of the new agile IS from the beginning. Once the habit is developed, users are likely to continue using the agile IS and explore new features when they become available.

Companies should also provide continuous organizational and technical support to users of agile IS, who are more sensitive to the costs than to the benefits of constant upgrades. Training sessions, maybe in the form of online lectures or software demonstration, can be provided every time there is a major release of new features. Technical support needs to be in place to ensure that when users have problems using the system, they have immediate access to technical assistance. Our results showed that a supportive infrastructure can promote users' willingness to not only continue using the agile IS but also to use the new features, which is critical to the ultimate success of agile IS.

Meanwhile, the benefits of users' adoption considerations, such as satisfaction and PU, play a less significant role in the context of agile IS than they do in the traditional IS adoption context. While the provision of useful features that are satisfying to users can help to retain them, it is not enough to convince them to use future features. However, this does not mean that software developers should stop emphasizing the provision of useful and satisfying features. The results simply suggest that there are other drivers of users' willingness to use new features, such as avoiding dramatic changes to the user interface and providing adequate technical support.

It is also difficult to affect user acceptance of agile IS through SI. Social pressure to start using a whole new system is stronger than the pressure to use new features of an existing system. Practically, in the agile IS context, it is unrealistic to exert organizational influence on users every time there is a new release. In the company we conducted the study on, e-mails were sent to users and headlines were posted on the portal before each round of new release, encouraging users to use these new features. But still, SI did not increase users' intentions to continue using the system or to use future features. Companies and software developers should not count on SI alone to promote the usage of agile IS or of future features.

Finally, companies should note that there are variations in individual employee's level of inclination toward constant technological changes. Employees who have higher levels of personal innovativeness are risk takers and are more likely to try new technology. If companies can determine their employees' level of personal innovativeness (from a psychological test during their job application or from evidence of

being early adopters of prior new technologies, for example), then they can target the more innovative employees as early users of an agile system while providing training classes for those employees who are late movers in adopting new technology. As innovative people are more interested in using future upgrades, companies can also select them as user representatives to work with the development team. By prompting early adopters who are more innovative and supporting later adopters who need more assistance, companies can reap the benefits of agile IS.

Limitations and Future Research

This study has some limitations. First, a longitudinal study would be ideal to study the effect of changing environments (i.e., initial adoption, six months later, one year later) on the acceptance of agile IS. However, because of the privacy policy of Company X, we could not track users' identities, making a longitudinal study impossible. Still, taking a "snapshot" of the agile IS environment allows us to derive some insights into user acceptance of agile IS. Nevertheless, future studies can design longitudinal data collections to validate and extend our findings. Second, this study, as an initial investigation into the acceptance of agile IS, has explored a limited set of new variables in addition to variables identified in the traditional IS adoption context. While it allows us to build upon the rich knowledge in IS adoption research and examine its applicability in the agile context, future research can explore additional variables that are unique to the acceptance of agile IS, such as users' perceptions of the rate of change, users' expectations of future upgrades, or users' involvement with system development. In particular, user involvement may have significant effects on adoption intentions but we could not test it in our study because of the limited number of user representatives. Future research can include this factor, especially for relatively small systems where more users can be directly involved in the development process. Third, our study has examined the total changes to an agile system. Future research could conduct a feature-level analysis, as users may adopt different features at different rates. Finally, the current study was conducted in a voluntary use setting. It would be useful to compare the findings to users' perceptions of agile IS under a mandatory use setting.

There are other avenues for further research into the acceptance of agile IS from the users' perspective. For example, is the acceptance of agile IS different from nonagile IS in terms of initial adoption, intention to continue to use, and intention to use upgrades? While the current study investigated one example of agile IS and found that there are different sets of facilitators for intention to continue using agile IS and intention to use future upgrades of agile IS, future research could test the generalizability of our findings with other agile IS. Does the acceptance of agile IS vary by release? That is, would the results vary if the study was conducted after the first release, after the *n*-release, after one year, or after two to three years (when a steady state has been reached)? Knowledge about how the results vary at different stages of the life cycle of an agile system would help to inform about user acceptance of agile IS. What is the role of habit at the initial stage and during subsequent upgrades to agile IS? The role

of habit with an agile system after one year is likely to be different from its role at the initial stage. It will be interesting to investigate how habit evolves with the frequency and magnitude of use of the upgrades to the agile system. Does user acceptance of agile IS vary by type of upgrade (i.e., reactive versus proactive). Finally, does the frequency and magnitude of upgrades influence user acceptance of agile IS? In other words, is there an ideal level of frequency of upgrades or number of new features in each upgrade that users will find most comfortable?

Conclusions

THE QUESTIONS ABOUT USERS' INTENTIONS TO CONTINUE USING agile IS and to use future upgrades become even more important in an era of smaller applications (e.g., iPhone and iPad apps), SaaS, and almost everything Web-based (e.g., Google's continuous updates of their ubiquitous systems), among many others. In this study, we investigated user acceptance of agile IS with the tripartite model of attitude, the status quo and the omission bias theories, and the availability heuristic. We found that the cognitive, affective, and behavioral dimensions of attitude play different roles in determining users' intention to continue using agile IS and intention to use future features. While some of the variables from the traditional IS adoption theories are still salient in this new context, others play less significant roles. A notable unique driver of agile IS acceptance is users' level of comfort with the constant changes. Further, we identified the precursors of comfort, in particular the need to maintain consistency with the existing system layout and with user knowledge during the course of periodic upgrades. The present work fills a gap in the software engineering literature and contributes to the understanding of user acceptance of agile IS, which are becoming a mainstay of companies' IT portfolio in a fast-changing business environment.

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NOTES

1. The periodic upgrades of agile IS are different from corrective or adaptive maintenance upgrades, which refer to *reactive* modifications of a software to correct problems discovered after delivery and to keep the software usable when a condition changes. Reactive modifications are performed after the full-featured software is put into operation, with emphasis given to revisiting the earlier developmental phases and correcting their associated oversights and mistakes [73]. Although adaptive maintenance may also entail addition of new features, it is typically performed under conditions that a change is typically unavoidable if the program is to be kept operable [72]. This is in contrast with the upgrades of agile IS that are *proactive* in nature, with the provision of new capabilities as their main goal, which leads to continuous upgrades and expansion of the software.

2. The Agile Manifesto [8] emphasizes the role of user involvement in the development process. User representatives are expected to work closely with the development team through collocation. In reality, user representatives have other job responsibilities, making collocation

difficult to achieve [31]. Further, collocation with the development team is more realistic for small-scale rather than for medium- to large-scale projects. When there are hundreds or even thousands of potential users of a system, only a small number can be involved directly. In such cases, surveys/e-mails are often used to solicit additional feedback from the majority of users, who serve as recipients and observers of agile IS rather than active participants in its development process.

3. We do not propose a causal relationship between the dependent variables as the aim of this research is to examine the antecedents of each intention variable. Also, the direction of the causal relationship between the dependent variables is unclear. On the one hand, the intention to continue using an agile IS is a prerequisite for intention to use future upgrades, as a user will not be able to explore new features if the user discontinued using the system. On the other hand, a user may only be interested in using the existing features and not the new features, or only be interested in using the new features and not the existing features.

4. This scenario is similar to other experiences we encountered in everyday life. For example, a man who has been using Gillette razors is told that Gillette has a new line of product and is asked whether he would be interested in trying the new product. If the man liked the performance of the previous products from Gillette, chances are he would be more willing to try the new product from Gillette.

5. We controlled for duration of usage and usage frequency in the structural model, but neither had a significant effect on the dependent variables and were therefore removed from further analysis.

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Appendix Table A1. Constructs and Measurement Items

Construct/ measurement items	Adapted from
Intention to continue using the agile IS	[10]
1. I intend to continue using <i>The System</i> * rather than discontinue its use.	
2. My intention is to continue using <i>The System</i> rather than using any alternative means.	
Intention to use future features	[88]
1. I predict that I will use the new features when they become available.	
2. I intend to use the new features when they become available.	
Satisfaction	[10]
What is your overall satisfaction with the upgrades of <i>The System</i> ?	
1. Very dissatisfied . . . Very satisfied	
2. Very unpleasant . . . Very pleasant	
3. Very frustrated . . . Very contented	
Confirmation	[10]
1. My overall experience with using the upgrades of <i>The System</i> was better than what I expected.	
2. The service level provided by the upgrades of <i>The System</i> was better than what I expected.	
Perceived usefulness	[88]
1. Using the upgrades of <i>The System</i> enables me to accomplish tasks more quickly.	
2. Using the upgrades of <i>The System</i> increases my productivity.	
3. Overall, the upgrades of <i>The System</i> are useful in my job.	
Perceived ease of use	[88]
1. Learning to use the upgrades of <i>The System</i> is easy for me.	
2. My interaction with the upgrades of <i>The System</i> is clear and understandable.	
3. I find the upgrades of <i>The System</i> easy to use.	
Social influence	[88]
1. My management promote the use of the upgrades of <i>The System</i> .	
2. My co-workers think that I should use the upgrades of <i>The System</i> .	
Facilitating conditions	[88]
1. I have the technical resources necessary to use the upgrades of <i>The System</i> .	
2. I have the knowledge necessary to use the upgrades of <i>The System</i> .	
Comfort with change	[79]
1. I feel comfortable with the changes resulting from the upgrades of <i>The System</i> .	
2. The continuous pace of upgrades of <i>The System</i> does not bother me.	
Consistency with user knowledge	[56]
1. In the upgrades, the use of buttons, radio buttons, and combo boxes is consistent with my understanding.	
2. In the upgrades, the data display is consistent with my usage conventions.	
Habit	[42]
1. Using <i>The System</i> has become a habit to me.	
2. Using <i>The System</i> has become automatic to me.	
Personal innovativeness	[1]
1. If I heard about a new information technology, I would look for ways to experiment with it.	
2. Among my peers, I am usually the first to try out new information technologies.	
3. I like to experiment with new information technologies.	

* The real name of the agile IS was substituted for *The System* in the questionnaire.